distant Universe and theoretical physicists who simulate the coalescence on supercomputers.

Box 10.1

Nonlinearity and Its Consequences

A quantity is called *linear* if its total size is the sum of its parts; otherwise it is *nonlinear.*

My family income is linear: It is the sum of my wife's salary and my own. The amount of money I have in my retirement fund is nonlinear: It is not the sum of all the contributions I have invested in the past; rather, it is far greater than that sum, because each contribution started earning interest when it was invested, and each bit of interest in turn earned interest of its own.

The volume of water flowing in a sewer pipe is linear: It is the sum of the contributions from all the homes that feed into the pipe. The volume of snow flowing in an avalanche is nonlinear: A tiny trickle of snow can trigger a whole mountainside of snow to start sliding.

Linear phenomena are simple, easy to analyze, easy to predict. Nonlinear phenomena are complex and hard to predict. Linear phenomena exhibit only a few types of behaviors; they are easy to categorize. Nonlinear phenomena exhibit great richness—a richness that scientists and engineers have only appreciated in recent years, as they have begun to confront a type of nonlinear behavior called *chaos.* (For a beautiful introduction to the concept of chaos see Gleick, 1987.)

When spacetime curvature is weak (as in the solar system), it is very nearly linear; for example, the tides on the Earth's oceans are the sum of the tides produced by the Moon's spacetime curvature (tidal gravity) and the tides produced by the Sun. By contrast, when spacetime curvature is strong (as in the big bang and near a black hole), Einstein's general relativistic laws of gravity predict that the curvature should be extremely nonlinear—among the most nonlinear phenomena in the Universe. However, as yet we possess almost no experimental or observational data to show us the effects of gravitational nonlinearity, and we are so inept at solving Einstein's equation that our solutions have taught us about the nonlinearity only in simple situations—for example, around a quiescent, spinning black hole.